

Sasumana Vinay Kumar

Senior AI ML Engineer

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SUMMARY

Innovative AI/ML Engineer with over 4.5 years of hands-on experience in Deep Learning, Generative AI, and scalable AI solutions, particularly in Video Analytics, Computer Vision, Action Recognition, and multimodal intelligence systems. Skilled in translating complex requirements into high-impact AI applications using cutting-edge architectures, including State-of-the-Art (SOTA), Large Vision-Language Models (LVLMs), and other foundation models. Proficient in designing and deploying robust end-to-end pipelines across diverse modalities such as video, audio, text, and multimodal data. Continuously evolving with the latest industry trends and emerging AI technologies, with a strong commitment to innovation, research-driven development, and delivering tangible business value through intelligent AI solutions.

EDUCATION

MASTER OF SCIENCE	December 2021 - December 2023
Artificial Intelligence and Machine Learning	
Liverpool John Moores University	England, United Kingdom
• Secured CGPA: 3.8/4	
BACHELOR OF TECHNOLOGY	July 2017 - August 2021
Computer Science and Engineering	
Sagi Ramakrishnam Raju Engineering College	Bhimavaram, Andhra Pradesh
• Secured CGPA: 8.88/10	

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer	May 2023 - Present
VideoVerse	Remote, IN

VideoVerse is dedicated to revolutionizing content creation through accessible, user-friendly video solutions empowered by AI and deep learning technologies.

Automated Real-Time Highlight Generation in Cricket Sports Broadcasting

- Build a multi-modal AI system for real-time highlight generation from .m3u8 live streams using **VideoMAEv2** with **FlashAttention**, **Whisper**, and **YOLO**
- Constructed parallel processing pipelines for audio analysis, scoreboard OCR, scene classification, and object detection, enabling **real-time tagging** of key cricket events such as **Wickets**, **Sixes**, **Fours**, **Centuries**, **Drop Catches**, and **Replays**, achieving **95% accuracy**
- Designed a scalable AI rule engine integrating LLMs (GPT, Gemini) for metadata generation, achieving **90%+ precision** in event detection and boosting automation efficiency for media workflows by over **95%**
- Architected a **Broadcast-Grade Single-Pass Delivery + Replay Detection and Meta-Tag Extraction pipeline** using **Gemini**, enforcing strict Pydantic JSON schema compliance for structured cricket event outputs, eliminating post-processing layers and reducing inference latency by **40%** while maintaining **90%+ precision** in automated metadata tagging for live streams

Key Movement Identification and Extraction in the Basketball Game.

- Composed a robust pipeline for key basketball game movement identification and extraction, integrating **YOLOv8** models with over 98% accuracy in ball and hoop detection
- Enforced advanced video classification algorithms, including **CNN-RNN** with **Multi-Headed Attention** and **Self-Supervised Contrastive** models, alongside **Custom VideoMAEv2** and **Zero-Shot Learning** models, enhancing pipeline efficiency by up to 95%
- Drove end-to-end **optimization** efforts to ensure seamless and **precise identification** of crucial movements, optimizing content production processes for enhanced **user experience** and **productivity**, and improving processing **speed by 2X**

Streamlined Analysis: Separating Live Feeds and Replay Segments in Sports Streaming.

- Engineered a **CNN-RNN** model to precisely distinguish transitions between **live** events and corresponding **replay** clips within sports streaming feeds
- Made an **Attention-Based Encoder** video classification model adept at categorizing video clips into distinct classes, including **live** events, **replay** segments, and **other** content, resulting in heightened segmentation accuracy
- Facilitated seamless integration of CNN-RNN and Attention-based encoder models for transition identification and video classification, achieving **97% efficiency** in segregating live feeds and replay segments in sports streaming

AI-Driven Logo Detection and Dynamic Brand Visibility Tracking

- Developed a scalable AI-based Logo Detection system enabling automatic brand tracking across sports broadcasts using both **GenAI** and in-house object detection models (**YOLOv11**)
- Deployed a dual-path solution using GCP Gemini 2.5 for instant **zero-shot** logo detection in live matches and transitioned to fine-tuned **YOLOv11** models per client, ensuring precision and cost efficiency
- Enabled sponsor reporting, clip filtering, and monetization workflows through **real-time logo analytics** across **1K+ brands**, supporting dynamic ad pricing and delivering scalable performance under tight tournament deadlines

AI-Assisted Vocal Preservation and Copyright-Free Music Replacement

- Formulated an AI pipeline to preserve vocals and replace background music in video content, leveraging **Demucs** for audio source separation and **MusicGen** for context-aware music generation
- Integrated **BPM detection** with Librosa and context-aware music generation via **MusicGen**, ensuring AI-composed tracks matched original tempo and mood with **100% vocal integrity**
- Devised a multi-modal background music removal framework using **SAM-Audio**, leveraging text, visual, and temporal span prompting to isolate dialogue from complex TV and live broadcast content
- Delivered an automated end-to-end solution for real-time audio replacement, reducing copyright risk by **>90%** and enabling **seamless remixing** of live and archived video content

SOFTWARE ENGINEER

September 2021 - May 2023

Tavant Technologies India Pvt. Ltd

Hyderabad, IN

Tavant Technologies excels in delivering impactful solutions and customer success, driven by passion, expertise, and dedication across AI, Cloud, Salesforce, Quality Engineering, and more.

Instance Segmentation and Pose Estimation in Video with YOLOv7 Models

- Executed instance segmentation and pose estimation in videos within the computer vision domain, utilizing **YOLOv7** models for robust performance.
- Spearheaded the integration of **YOLOv7** models to achieve a 93% efficiency in **Instance Segmentation** and subsequent **Pose Estimation** tasks, enhancing video analysis capabilities and facilitating advanced computer vision applications with high accuracy

Data Anonymization and PII Redaction using NLP and NER Techniques

- Implemented NLP and NER techniques to anonymize data and redact Personally-Identifying Information (**PII**) proactively
- **Directed** the integration of AI-driven redaction workflows into enterprise pipelines, enhancing **data security** and **confidentiality** while reducing manual review efforts by **80%**

TECHNICAL SKILLS

- **Machine Learning:** Statistical ML, Supervised, Unsupervised ML Algorithms, Feature Engineering.
- **Deep Learning:** ANN, CNN, RNN, LSTM, GRU, Attention Mechanisms, Contrastive learning, Self-Supervised Learning.
- **Computer Vision:** Video Analytics, Video Classification, Action Recognition, Transformers, Object Detection, GANs, Segmentation.
- **Generative AI (GenAI):** Prompt Engineering, Multimodal Reasoning, Audio & Music Generation, Text-to-Video & Text-to-Image Foundations, LLMs (GPT, Gemini), OCR + LLM Integration, Video Commentary Generation.
- **Agentic AI & LLM Systems:** ReAct Agents, RAG Architectures, Autonomous Workflows, MCP (Model Context Protocol), Memory-Augmented Systems.
- **NLP:** Lexical, Syntactic, and Semantic Processing, Attention-Based NMT.
- **DL Frameworks:** PyTorch, Keras, TensorFlow, Hugging Face Transformers.
- **Cloud Computing:** AWS, Azure.

ACCOMPLISHMENTS

- Completed Master's Thesis on **Self-Supervised Contrastive Learning for Video Understanding and Action Recognition**, exploring scalable video representation learning using unlabeled data for improved performance in downstream tasks like action recognition.
- Attended the University Innovation (**UI**) Fellows Silicon Valley Meetup 2022 at **Stanford University**, United States.
- **University Innovation Fellow** (Fall 2019 – Fall 2023), promoted to **University Innovation Guide** (Fall 2023 – Present) - Stanford University
- Wrote a [Research Paper](#) entitled Comparative Study of Machine Learning Models to Classify Gene Variants of ClinVar.
- Wrote a [Survey Paper](#) entitled Machine Learning: Concepts, Algorithms, and Applications.

CERTIFICATIONS

- Advanced Certification in **Machine Learning and Cloud** from IIT Madras.
- ACM India Winter School 2021-22 on **Quantum Computing** Hosted by IIT Madras and TCS Research.
- Fundamentals of Deep Learning for **Computer Vision** from NVIDIA.